Read-in action in mixed Hermite space

Two-layer erf nets with a nondiagonal He1–He3 block J

Notes for discussion (notation as in Lipshitz MSc thesis)

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Abstract

These notes specialize the thesis notation to a *two-layer* erf network and keep the teacher-aligned Hermite block of the first-layer kernel *nondiagonal*. The thesis mean-field treatment often works in a basis where that block is diagonal; here we keep the off-diagonal matrix element J_{13} explicit. We (i) write the posterior in the mixed He1–He3 mode space, (ii) integrate the readout to obtain a quadratic amplitude matrix G acting on Stein features $c(w)$, (iii) obtain the scalar read-in action $V(w)$ along the teacher axis, and (iv) perform a Ginzburg–Landau expansion that identifies which invariant order parameters control the inner (center) versus outer well transitions, and how ε enters. All a_0 dependence is written explicitly: the cavity resolvent is $(a_0 J + \frac{\kappa}{P} I)^{-1}$; we never introduce auxiliary primed scales.

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1 Setup and notation

We work with a two-layer network with erf activations,

$$f_\Theta(x) = \sum_{i=1}^{N} a_i \, \Phi\big(w_i \cdot x\big), \qquad \Phi = \mathrm{erf}, \tag{1}$$

with read-in rows $w_i \in \mathbb{R}^d$ collected in $W \in \mathbb{R}^{N\times d}$ and readout $a \in \mathbb{R}^N$. (The three-layer objects B, h_k, ν_k of the thesis are absent.)

Data and target. Inputs $x \sim \mathcal{N}(0, I_d)$. The target lives in the span of the first two odd probabilists' Hermite polynomials along a unit teacher direction v_* ($\|v_*\| = 1$):

$$y(x) = \mathrm{He}_1(v_* \cdot x) + \varepsilon\, \mathrm{He}_3(v_* \cdot x), \tag{2}$$

with $\mathrm{He}_1(z) = z$ and $\mathrm{He}_3(z) = \frac{1}{\sqrt{6}}(z^3 - 3z)$ as in the thesis. We write Hermite feature maps $\varphi_1(x) = \mathrm{He}_1(v_* \cdot x)$ and $\varphi_3(x) = \mathrm{He}_3(v_* \cdot x)$.

Inner products. Dataset / population averages use the input measure $d\mu_x$:

$$\langle h \mid g \rangle_x := \int d\mu_x\, h(x)\, g(x), \qquad \|h\|_x^2 := \langle h \mid h \rangle_x. \tag{3}$$

Operator kernels act as $\big(Kg\big)(x) = \int d\mu_{x'}\, K(x, x') g(x')$.

Hyperparameters.

- P: number of training samples;
- κ: noise scale in the data term (Langevin temperature $T \equiv 2\kappa$);
- χ: train / loss scale in the data term;
- a_0: readout prior scale, so that the readout prior variance per neuron is $\sigma_A^2 = \dfrac{a_0}{N\chi}$;
- s_0: read-in prior variance scale (usually $s_0 = 1$).

Bayesian action.

$$S[a, W] = \frac{N\chi}{2a_0} \|a\|^2 + \frac{d}{2s_0} \|W\|^2 + \frac{P\chi}{2\kappa} \big\|f - y\big\|_x^2. \tag{4}$$

The posterior is $p(a, W) \propto e^{-S[a,W]}$. The Gaussian readout measure implied by the first term is $a \sim \mathcal{N}\big(0, \frac{a_0}{N\chi} I\big)$.

First-layer kernel J.

$$J(x, x') := \mathbb{E}_{W \sim p(W|D)} \Big[\tfrac{1}{N} \sum_{i=1}^{N} \Phi(w_i \cdot x)\, \Phi(w_i \cdot x') \Big]. \tag{5}$$

In the large-d Hermite basis aligned with v_*, it is enough for the teacher (2) to retain the 2×2 block of Rayleigh quotients

$$J_{mn} := \langle \varphi_m \mid J \mid \varphi_n \rangle_x, \qquad m, n \in \{1, 3\}, \tag{6}$$

i.e. the finite matrix

$$J = \begin{pmatrix} J_{11} & J_{13} \\ J_{13} & J_{33} \end{pmatrix}. \tag{7}$$

The thesis often assumes a basis in which $J_{13} = 0$. We do *not*: mixed space means J_{13} is retained.

Remark 1 (Perp modes)**.** *Modes orthogonal to v_* remain (to leading large-d order) unadapted and are absorbed into an effective floor*

$$T = 1 + 2\, \frac{d - 1}{d}\, s_0 \tag{8}$$

in the Stein formulae below; they do not enlarge the 2×2 teacher block.

2 Posterior in mixed (He1–He3) mode space

Project the network output and the target onto $\mathrm{span}\{\varphi_1, \varphi_3\}$. Write mode coefficients

$$f_m := \langle \varphi_m \mid f \rangle_x, \qquad y_m := \langle \varphi_m \mid y \rangle_x, \tag{9}$$

so that, for the target (2),

$$y = \begin{pmatrix} y_1 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix} \tag{10}$$

(up to the conventional normalization of He_3; absolute scales can be absorbed into ε).

Restricted to this subspace, the free (Gaussian) mode-space action with kernel J is

$$S_{\mathrm{mix}}[f] = \frac{1}{2}\, f^\top J^{-1} f + \frac{P\chi}{2\kappa}\, \|f - y\|^2, \tag{11}$$

where $f = (f_1, f_3)^\top$ and $\|f - y\|^2 = (f-y)^\top (f-y)$.

a_0 in the resolvent. After integrating the Gaussian readout with variance $a_0/(N\chi)$, the cavity field in mode space is defined by

$$\boxed{v := \Big(a_0 J + \frac{\kappa}{P}\, I_2\Big)^{-1} y} = \Big(a_0 J + \frac{\kappa}{P}\, I_2\Big)^{-1} \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}. \tag{12}$$

Write $R := a_0 J + \frac{\kappa}{P} I$ and $\Delta := \det R$. Then

$$v_1 = \frac{1}{\Delta}\Big[\big(a_0 J_{33} + \tfrac{\kappa}{P}\big) - a_0 J_{13}\, \varepsilon\Big], \tag{13}$$

$$v_3 = \frac{1}{\Delta}\Big[-\, a_0 J_{13} + \big(a_0 J_{11} + \tfrac{\kappa}{P}\big)\varepsilon\Big]. \tag{14}$$

Both components depend on ε and on the mixing element J_{13}. If $J_{13} = 0$,

$$v_1 = \Big(a_0 J_{11} + \tfrac{\kappa}{P}\Big)^{-1}, \qquad v_3 = \varepsilon \Big(a_0 J_{33} + \tfrac{\kappa}{P}\Big)^{-1}. \tag{15}$$

Learnability. The mode-space posterior mean is

$$\bar{f} = a_0 J\, v = a_0 J \Big(a_0 J + \tfrac{\kappa}{P} I\Big)^{-1} y. \tag{16}$$

In the eigenbasis of J with eigenvalues λ_m,

$$\tilde{L}_m = \frac{a_0 \lambda_m}{a_0 \lambda_m + \kappa/P}. \tag{17}$$

Learnability is controlled by the spectrum of J versus the ridge κ/P (with a_0 weighting J), not by which wells of the eventual read-in potential are occupied.

3 Amplitude matrix G

Integrating $a \sim \mathcal{N}\big(0, \frac{a_0}{N\chi} I\big)$ replaces the data term by a quadratic form in the first-layer features. In the He1–He3 block the amplitude matrix is

$$\boxed{G = \frac{a_0}{N\chi}\left[-\, \chi^2\, \delta\, v v^\top + \frac{\chi}{a_0}\, J^{-1}\right]}, \tag{18}$$

with v from (12) and $\delta = 1$ for standard square loss. Equivalently,

$$G = \frac{a_0}{N\chi}\big(-\chi^2\delta\, vv^\top\big) + \frac{1}{N}\, J^{-1}. \tag{19}$$

The first (discrepancy) term is negative semidefinite and rank-one along v; the second (Onsager) term is positive definite when $J \succ 0$.

Writing $G = G_{\text{disc}} + G_{\text{ons}}$,

$$\tfrac{1}{2}\, c^\top G_{\text{disc}}\, c = -\frac{a_0\chi}{2N}\,\delta\,(v_1c_1 + v_3c_3)^2, \qquad \tfrac{1}{2}\, c^\top G_{\text{ons}}\, c = \frac{1}{2N}\, c^\top J^{-1} c. \tag{20}$$

4 Stein features and the scalar read-in action

Reduce each read-in row to the teacher coordinate $w := w_i \cdot v_*$. Stein evaluations of $\langle \Phi \circ w \mid \mathrm{He}_n \circ v_* \rangle_x$ define scalar features $c(w) = (c_1(w), c_3(w))$:

$$c_1(w) = \sqrt{\frac{4}{\pi}}\, \frac{w}{(T + 2w^2)^{1/2}}, \tag{21}$$

$$c_3(w) = -\sqrt{\frac{8}{3\pi}}\, \frac{w^3}{(T + 2w^2)^{3/2}}, \tag{22}$$

normalized so that, in a Laplace / mixture ansatz for $p(w)$,

$$J \approx \mathbb{E}_{w\sim p(w)}\big[c(w)c(w)^\top\big]. \tag{23}$$

(We do not assume a sign for c_3 beyond the conventional overall phase in (22).)

The effective single-neuron read-in action is

$$\boxed{V(w) = \frac{d}{2s_0}\, w^2 + \tfrac{1}{2}\, c(w)^\top G\, c(w)} \tag{24}$$

with G from (18). Self-consistency closes by rebuilding J from the wells of V: locate wells $\{\mu_k, \sigma_k\}$ of V, assign Laplace mixture weights $\pi_k \propto m_k \sigma_k e^{-V(\mu_k)}$ ($m_k = 2$ for a $\pm\mu$ pair), and set $J = \sum_k \pi_k\, \mathbb{E}_{w\sim\mathcal{N}(\mu_k,\sigma_k^2)}[c(w)c(w)^\top]$.

Where ε enters. With $R = a_0 J + \frac{\kappa}{P} I$,

$$v_1c_1 + v_3c_3 = \underbrace{\big(R_{11}^{-1}c_1 + R_{21}^{-1}c_3\big)}_{\phi_1(c)} \cdot 1 + \underbrace{\big(R_{12}^{-1}c_1 + R_{22}^{-1}c_3\big)}_{\phi_\varepsilon(c)} \cdot \varepsilon. \tag{25}$$

Hence the discrepancy contribution is proportional to

$$-\big(\phi_1(c) + \varepsilon\,\phi_\varepsilon(c)\big)^2 = -\phi_1^2 - \varepsilon^2\phi_\varepsilon^2 - 2\varepsilon\,\phi_1\phi_\varepsilon. \tag{26}$$

Only the last term flips under $\varepsilon \to -\varepsilon$. The Onsager piece $\frac{1}{2N}\, c^\top J^{-1} c$ depends on ε only indirectly through the self-consistent J.

5 Ginzburg–Landau treatment after stripping d

Define the d-stripped potential

$$\tilde{v}(w) := \frac{2s_0}{d}\, V(w) = w^2 + \frac{s_0}{d}\, c(w)^\top G\, c(w). \tag{27}$$

Phase structure of V coincides with that of $\tilde{v}$.

From (18), the dimensionless strength of the discrepancy channel is set by

$$g := \frac{s_0}{d} \cdot \frac{a_0 \chi}{N} = \frac{s_0\, a_0\, \chi}{N\, d}, \tag{28}$$

while the ridge scale that enters v is κ/P (together with the weighting a_0 of J in $R = a_0 J + \frac{\kappa}{P} I$). An invariant training ray holds combinations such as $(g, \kappa/(a_0 P))$ fixed while dialling ε, or scans P at fixed $(a_0, \kappa, \chi, N, d)$.

5.1 Small-w expansion (inner / center transition)

Near $w = 0$,

$$c_1(w) = \alpha_1 w + O(w^3), \qquad c_3(w) = \alpha_3 w^3 + O(w^5), \tag{29}$$

with α_1, α_3 fixed by (21)–(22) and T. Then

$$(v_1 c_1 + v_3 c_3)^2 = (v_1\alpha_1)^2 w^2 + 2 v_1 v_3 \alpha_1 \alpha_3\, w^4 + (v_3\alpha_3)^2 w^6 + \cdots, \tag{30}$$

and (discrepancy-dominated; Onsager absorbed into the coefficients)

$$\tilde{v}(w) = a_2 w^2 + a_4 w^4 + a_6 w^6 + \cdots, \tag{31}$$

with

$$a_2 = 1 - C\,(v_1\alpha_1)^2 + \cdots, \tag{32}$$

$$a_4 = -C\,(2 v_1 v_3 \alpha_1 \alpha_3) + \cdots, \tag{33}$$

$$a_6 = -C\,(v_3\alpha_3)^2 + \cdots, \tag{34}$$

and $C \propto g\,\chi\,\delta = \dfrac{s_0 a_0 \chi^2 \delta}{N d} > 0$ (up to the precise factor from (20)).

Invariant order parameters. Define cavity loads

$$\psi_1 := \sqrt{C}\, v_1, \qquad \psi_3 := \sqrt{C}\, v_3. \tag{35}$$

Then (a_2, a_4, a_6) are functions of $(\psi_1^2,\ \psi_1\psi_3,\ \psi_3^2)$. From (13)–(14),

$$\psi_1\psi_3 \ \propto\ v_1 v_3 = (\text{terms independent of } \varepsilon) + (\text{terms linear in } \varepsilon) + (\text{terms in } \varepsilon^2), \tag{36}$$

so sign(ε) moves a_4 at fixed $|\varepsilon|$ through the cross invariant $\psi_1\psi_3$.

Landau diagnostics.

Phenomenon	Condition on (a_2, a_4, a_6)	Controlling invariants
center spinodal	$a_2 = 0$, $a_4 > 0$	ψ_1^2
inner / central wells	$a_2 > 0$, $a_4 < 0$, $a_6 > 0$	$\psi_1\psi_3$ (He1–ε cross)
outer wells	large-$\lvert w\rvert$ / finite-μ analysis	$\lvert\psi_3\rvert$, J_{33}, κ/P, a_0
learnability	$\tilde{L}_m = \dfrac{a_0\lambda_m}{a_0\lambda_m + \kappa/P}$	spectrum of J vs κ/P

With $\varepsilon > 0$ and lab mixing, the cross term that feeds a_4 leaves the **inner/center** reorganization live, while the cubic channel $|\psi_3|$ opens **outer** minima at large $|w|$—two landscape transitions. Flipping $\varepsilon \to -\varepsilon$ flips only a_4's sensitive piece and can kill the inner transition, leaving a single landscape transition tied to the outer / learning channel.

5.2 Outer transition

The small-w series cannot see outer wells. One either matches the large-$|w|$ asymptotics of $\tilde{v}(w)$, dominated by $-C(v_3 c_3(w))^2$, or introduces a finite-well order parameter $m = \mu^2$ and expands an effective potential $U(m)$. Criticality is a condition on $\big(|\psi_3|,\, J_{33},\, \kappa/P,\, a_0\big)$, not on a_2.

5.3 Protocol to map the phase diagram

1. Fix an invariant ray: strip d; hold g and $\kappa/(a_0 P)$ (or equivalent).
2. Solve self-consistent $J \Rightarrow v = (a_0 J + \frac{\kappa}{P} I)^{-1} y \Rightarrow (\psi_1, \psi_3)$.
3. Compute (a_2, a_4, a_6) from the small-w expansion of $\tilde{v}$.
4. Draw boundaries in $(\psi_1^2, \psi_1\psi_3)$ or (a_2, a_4).
5. Overlay the outer boundary from a large-w / finite-μ condition.
6. Scan sign(ε) at fixed $|\varepsilon|$: this moves a_4 and diagnoses whether the inner transition is physical or a mixing artifact.

6 Summary

- Mixed-space posterior: keep the full 2×2 block J including J_{13}.
- Cavity field: $v = \left(a_0 J + \dfrac{\kappa}{P} I\right)^{-1} \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}$.
- Amplitude: $G = \dfrac{a_0}{N\chi}\Big[-\chi^2 \delta\, v v^\top + \dfrac{\chi}{a_0}\, J^{-1}\Big]$.
- Read-in action: $V(w) = \dfrac{d}{2 s_0} w^2 + \frac{1}{2} c(w)^\top G\, c(w)$.
- ε enters through $-\big(\phi_1 + \varepsilon \phi_\varepsilon\big)^2$; only $2\varepsilon\phi_1\phi_\varepsilon$ flips with sign(ε).
- After stripping d, Landau coefficients (a_2, a_4, a_6) are controlled by (ψ_1, ψ_3); inner $\leftrightarrow a_4 \leftrightarrow \psi_1\psi_3$; outer $\leftrightarrow |\psi_3|$; learning $\leftrightarrow$ spectrum of J versus κ/P (with weight a_0).

These notes are intended for discussion with the thesis advisor; they extend the diagonal-J treatment of the MSc thesis to mixed He1–He3 space for the two-layer read-in problem.